MODELLING PETROL PRICE OF USA USING THE TIME SERIES MODEL

STAT 358: Time Series Analysis

GROUP 12

MINI PROJECT

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ABSTRACT

The study describes variations and predictions. The petrol prices increased from the earlier weeks and subsequently declined. The Box-Jenkins methodology was used. The highest petrol price (142.7) and the lowest (74.21) were recorded around 4/16/2012 and 7/14/2003 respectively. The series was not stationary. The model for the change in petrol price is (ARIMA (1,1,1) (0,0,1) [52]). The petrol price is expected to stay around 61% and decline to 60.97%.

INTRODUCTION

Petrol is a transparent flammable liquid, derived from crude oil through a refinery process and mainly used primarily as a fuel in most spark-ignited internal combustion engines. It is an important liquid which affects the economy of a country by influencing general fluctuation in cost of goods and services of a country. Thereby Affecting the cost of living. Due to the direct relationship between petrol prices and the cost of living, it is important for the government to understand the waves in petrol prices in order to make sound policies. This study looked at the weekly USA petrol prices from 9th June 2003 to 20th April 2020. The purpose of the study was to examine the variations in the data, obtain a model and use it to predict for the next 30 weeks.

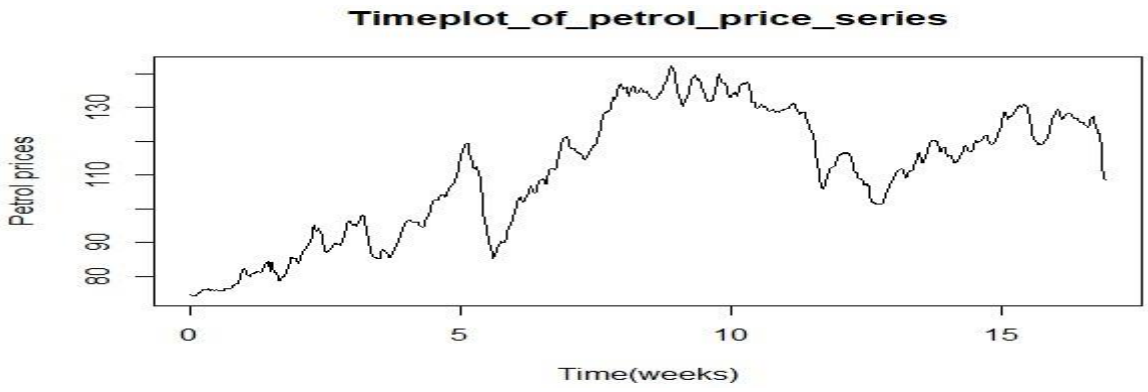
METHODOLOGY

To achieve the aim of this study a secondary data on the petrol price of the USA from 2003 to 2020 was used. The analysis of the data was done in R and the Box-Jenkins methodology was used. The time plot was used to determine the variations in the petrol prices. A histogram of the data was viewed to determine the most frequent petrol prices. The summary statistics was obtained. To check the stationarity of the time series, the plot of the acf and pacf of the series were obtained in addition to stationarity tests (kpss and adf). In the second and third stage of the methodology, the automatic search function (auto.arima()) was used to obtain a model for the data. The model was then used to forecast the petrol prices in the next 30 weeks.

RESULTS AND ANALYSIS

Model Identification Fig 1.1

```
>PetrolP1=read.csv(file.choose()) ; PetrolPrice=PetrolP1[,2];PetrolPrice.ts=ts(PetrolPrice, start
=6/9/2003,frequency= 52); plot.ts(PetrolPrice.ts, main = "Timeplot_of_petrol_price_series",ylab = "Petrol prices",
xlab = "Time(weeks)")
```



From **Figure 1.1**, petrol prices increased steadily from 2003 reaching the highest peak of around $100 somewhere around late 2007. Due to the oil shock of 2007-2008 across all over the world, petrol prices increased rapidly to the maximum in 2012. The prices remained at the study rate afterwards and started to decline in early 2020

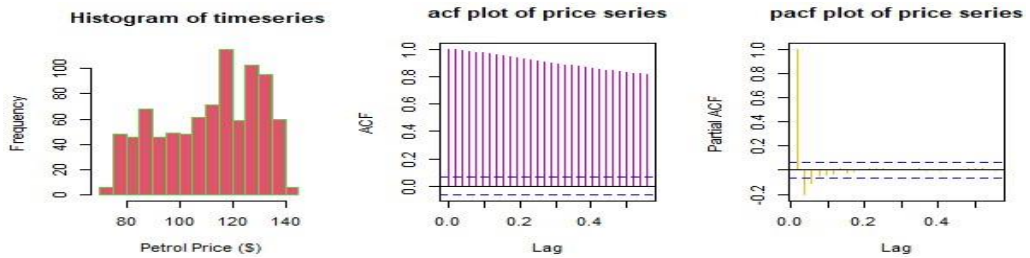
Table 1.1 Summary Statistics for Petrol Prices.

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	Sd
74.21	95.31	114.78	111.11	127.92	142.17	18.5882

> summary(PetrolPrice.ts); sd(PetrolPrice.ts)

From **Table 1.1**, the lowest petrol price was $74.21 and average price is 111.11 with standard deviation of 18.5882.

Figure 1.2



> hist(PetrolPrice.ts, main = "Histogram of timeseries", xlab = " Petrol Price" , col = 2, border =3); >acf(PetrolPrice.ts, main = "acf plot of price series", col = 6); > pacf(PetrolPrice.ts, main = "pacf plot of price series", col = 7)

Figure 1.2 above show histogram, acf and pacf correlograms; Over the Period, USA recorded prices of between $70 to 110 most often and recorded the least prices between 110 to 140. The acf plot of the rate decays slowly, suggesting that, the time series is not stationary. The pacf decays exponentially. To confirm this suggestion, stationary tests were done and result shown in **table 1.2**.

Table 1.2 Stationarity test Output

Test	P-value	Hypothesis
kpss	0.01	H_0: it is stationary, Ha: it is not stationary
adf	0.6422	H_0: it is not stationary, Ha: it is stationary

> kpss.test(PetrolPrice.ts ,null="Trend"); > adf.test(PetrolPrice.ts); ndiffs(PetrolPrice.ts)

Since alpha=0.05, we reject Ho and fail to reject Ho in the kpss and adf test respectively. The ndiffs function suggested number of differencing to achieve stationarity as 1. This means the value of d in the ARIMA(p,d,q) is expected to 1.

Model Identification and diagnostics.

Table 1.3 Estimated Model.

Model	**ar1**	**ma1**	**sma1**
Estimates	0.6927	-0.1057	0.0672
s.e.	0.0386	0.0537	0.0356

> auto.arima(PetrolPrice.ts); PetrolPriceArima = arima(PetrolPrice.ts, order = c(1,1,1), seasonal = list(order= c(0,0,1),period = 52))

The model is: ARIMA(1,1,1)(0,0,1)[52] with the coefficients and standard errors shown in **Table 1.3**. There a seasonal component, even though it is not of great concern to this study.

Forecasting

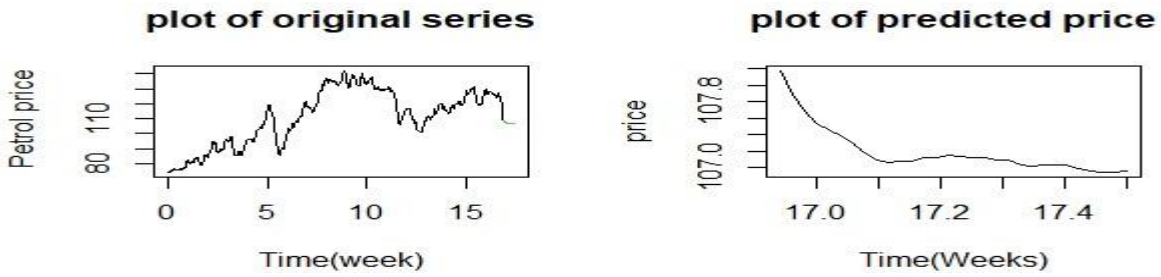
Predicted values for the next 30 weeks.

108.18 107.90 107.68 107.54 107.46 107.39 107.30 107.18 107.09 107.06 107.07 107.08 107.12 107.13 107.15 107.14 107.13 107.13 107.10 107.09 107.03 107.03 107.04 107.03 107.03 106.99 106.96 106.94 106.95 106.96

>PetrolPrice.pred = predict(PetrolPriceArima, n.ahead = 30); PetrolPrice.pred$pred

From the predicted values the Petrol Price is expected to decrease to around $107.90 and further decrease to $106.94. It indicates that the US Petrol Price is likely to decrease in the second quarter of 2020. the government must ensure that these reductions are realized by putting policies in place to handle shocks that will lead to surge in prices. This decline may be attributed to the impact of covid-19 pandemic which led to the global lockdown and limited movement of people to places requiring the use of automobiles.

FIGURE 1.3



> ts.plot(PetrolPrice.ts, PetrolPrice.pred$pred, col = c(1,3,2), lty = c(1,1,2,3), main = "plot of original series", ylab = " Petrol price", xlab = "Time(week)") ; > ts.plot(PetrolPrice.pred$pred, col= c(1,3,2), lty=c(1,1,2), main = "plot of predicted price", ylab = "price", xlab = "Time(Weeks)")

From **Figure 1.3** the Petrol Price for next 30 weeks is decreasing which implies that, the US Petrol Price is not going to be constant but rather decreases from around $107.90 to $106.96.

CONCLUSION

The Petrol Price of the USA has over the years gone through changes. The Petrol Price is expected to decrease to $106.96 by the end of 2020, which will happen as a result of the impact of the COVID-19. It is advisable that the Government puts measures in place to avoid unexcepted huge price changes (increases).